Decision Trees and Random Forests

DAEN 300 - Fall 2026

- Explain what a decision tree and random forest do;
- Spot a tree that is learning noise;
- Compare models fairly and make a recommendation.



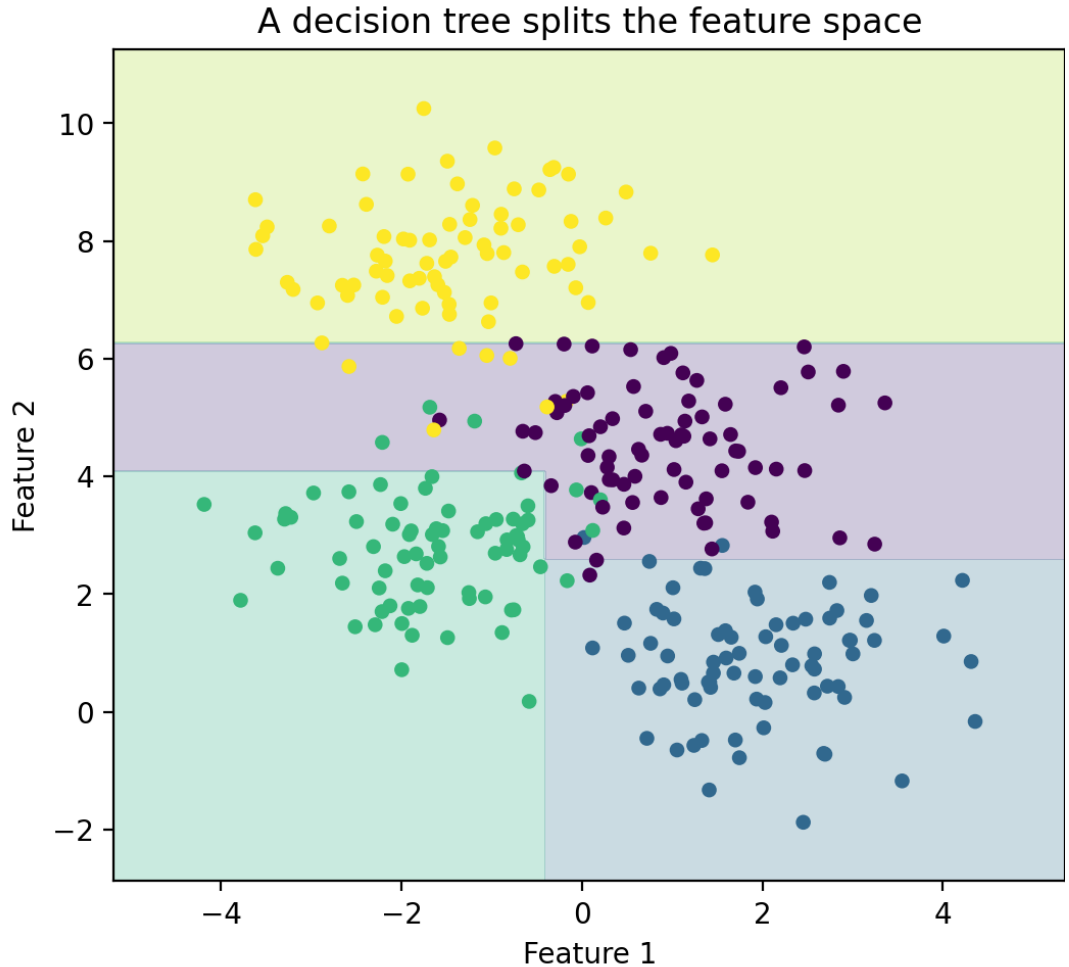
Would you trust this model?

A decision tree is nearly perfect on its **training data**.
But its prediction regions look jagged and unusually narrow.

- Is that a good sign, a warning sign, or both?
- What would you check before trusting it with new data?

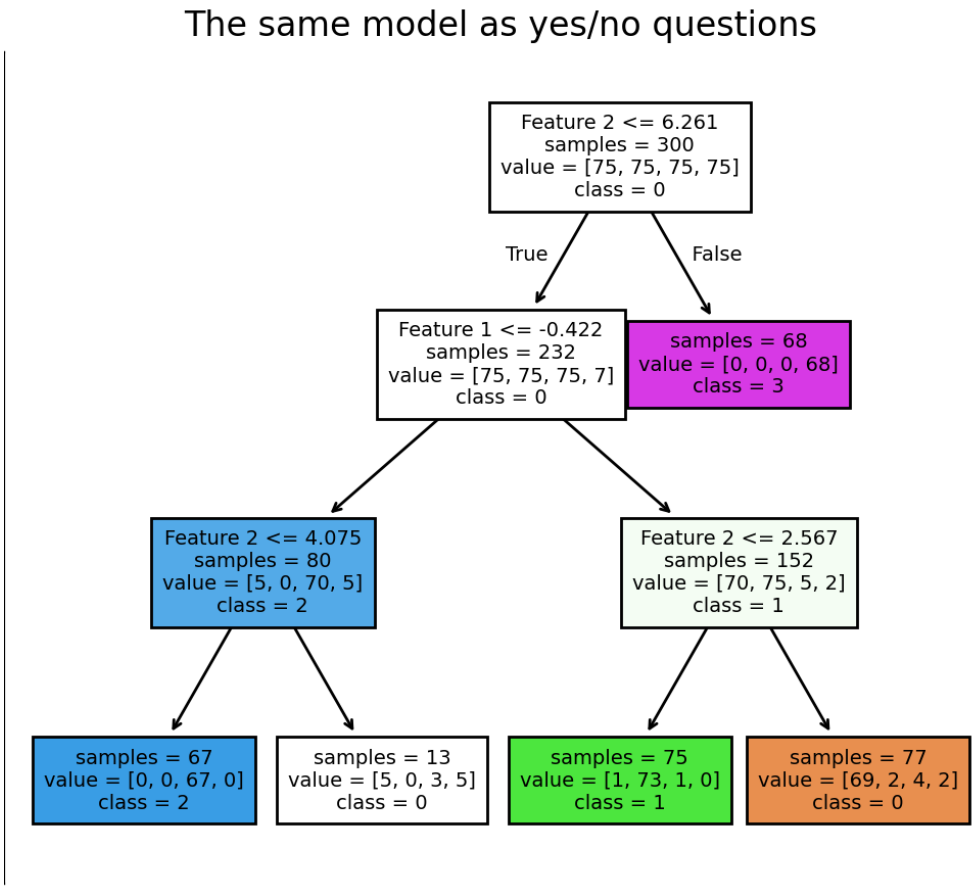
Think for 30 seconds, then compare answers with a partner (2 min).

A tree divides the data into regions



Each colored area is a prediction.
Every straight boundary comes from a simple feature threshold.

The same model is a set of questions



Follow a path from top to bottom: the final leaf tells us the predicted class.

Why deeper is not always better

A shallow tree learns broad patterns. A very deep tree can keep splitting until it fits small quirks in the training sample.

- **Underfitting:** misses important patterns.
- **Overfitting:** learns noise that may not repeat.
- **Generalization:** works well on new observations.

The model with the highest training score is not always the best model for new data.

Try different tree depths

```
from sklearn.datasets import make_blobs
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

X, y = make_blobs(n_samples=300, centers=4,
                  random_state=0, cluster_std=1.0)
Xtr, Xte, ytr, yte = train_test_split(
    X, y, test_size=.30, stratify=y, random_state=42)

for depth in [2, 5, None]:
    tree = DecisionTreeClassifier(
        max_depth=depth, random_state=0)
    tree.fit(Xtr, ytr)
    print(depth, tree.score(Xtr, ytr),
          tree.score(Xte, yte))
```

What did the depth experiment show?

Compare the training and test accuracy for three tree depths.

- Which tree had the strongest training score?
- Did it also have the strongest test score?
- Which model would you trust more on unseen data?

Write a two-sentence interpretation with your partner (3 min).

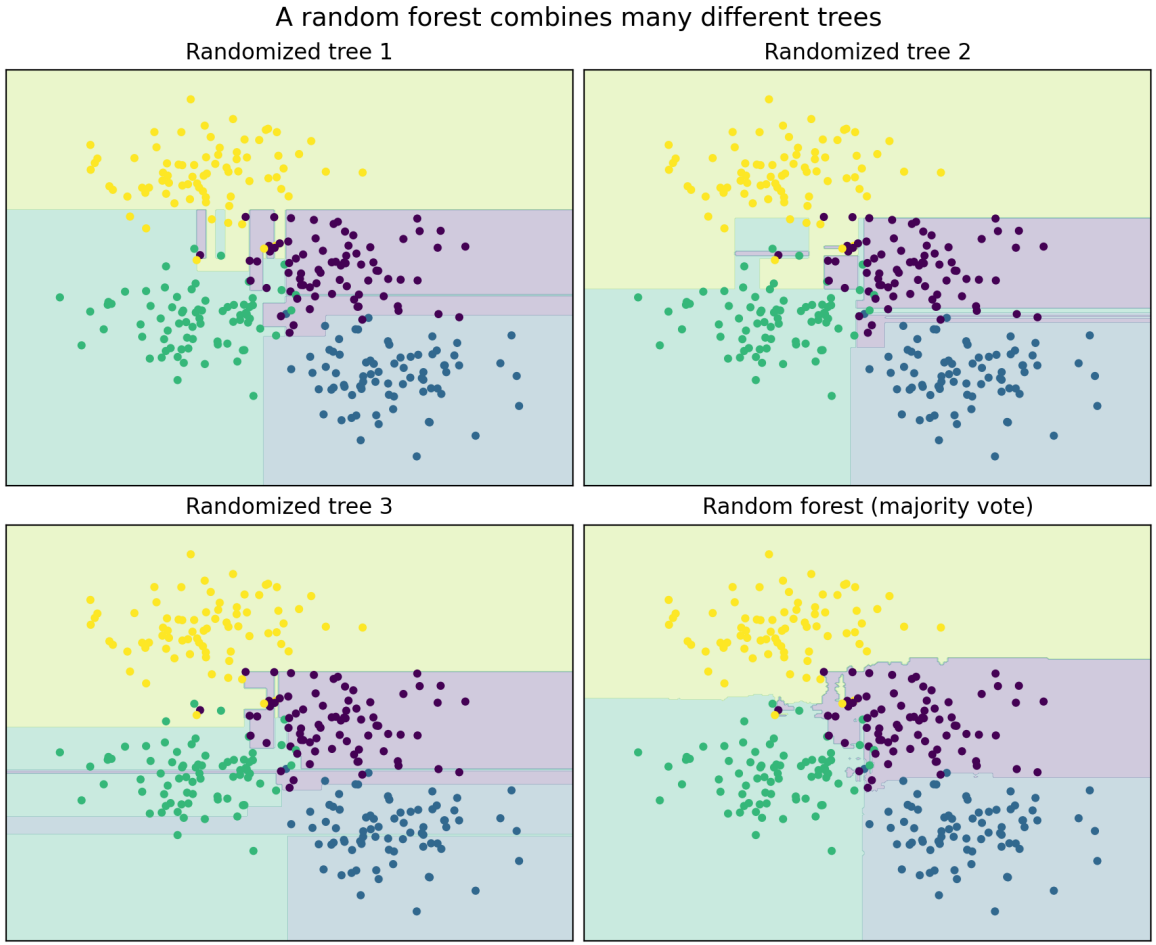
How could we make trees more reliable?

Imagine asking many slightly different decision trees to classify the same new point.

Would their combined answer be **more stable**, **less stable**, or **about the same** as one tree?

> Make a prediction and explain your reasoning.
> We will check it next.

One forest, many different boundaries



The trees disagree in some places.
Their combined prediction tends to be less sensitive to individual quirks.

What changes from tree to forest?

A **random forest** trains many decision trees and combines their outputs.

- Each tree sees a resampled training set (**bagging**).
- At each split, it considers a random subset of features.
- In classification, the forest combines class votes.
- In regression, it averages numeric predictions.

The useful ingredient is diversity:
different trees make different mistakes.

Let the trees vote

A B A A B

Five trees classify the same observation. What does the forest predict?

- Which label wins the vote?
- What share of trees support that label?
- Why would five identical trees be less useful?

The vote is A (3 out of 5 trees).
Vote share is not necessarily a calibrated probability.

Compare a tree and a forest

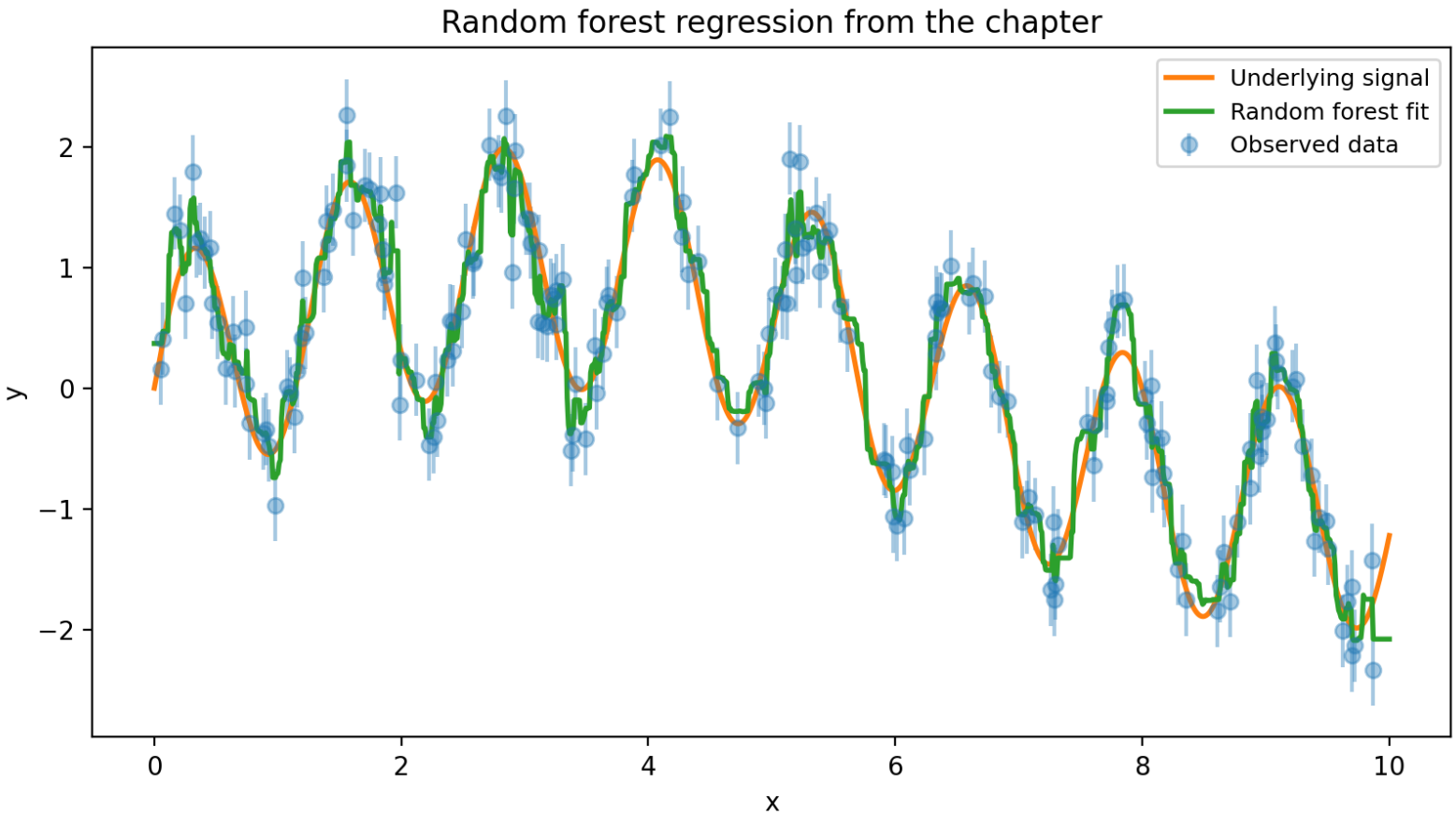
```
from sklearn.ensemble import RandomForestClassifier

models = {
    "tree": DecisionTreeClassifier(random_state=0),
    "forest": RandomForestClassifier(
        n_estimators=100, random_state=0)
}

for name, model in models.items():
    model.fit(Xtr, ytr)
    print(name, "train:", model.score(Xtr, ytr),
          "test:", model.score(Xte, yte))
```

Which model generalizes better on this split?
Do not assume the forest must win.

Forests can predict numbers, too



In the handbook's two-frequency example,
a forest follows a noisy nonlinear signal without being told its formula.

What does regression teach us?

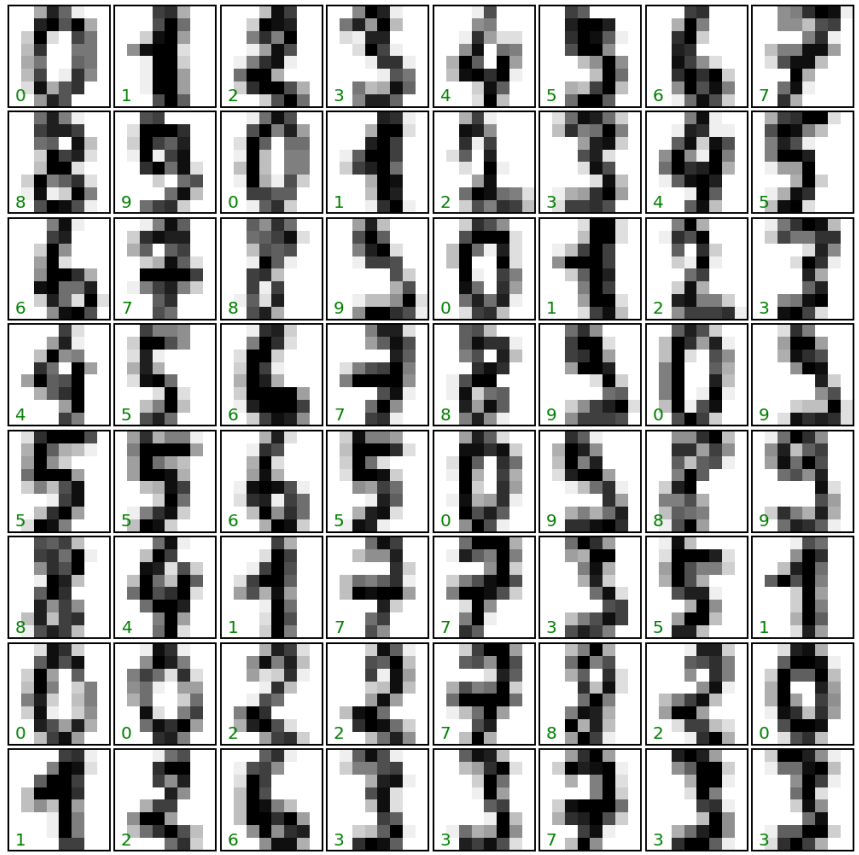
The chapter uses the signal

$$y = \sin(5x) + \sin(0.5x) + \varepsilon.$$

The forest can learn a flexible approximation from examples,
but its fitted curve can be jagged.

- Classification forests combine votes.
- Regression forests average predictions.
- Neither automatically discovers the underlying scientific equation.

A new task: recognize handwritten digits



Each image has 8×8 pixels (64 features).
Can the forest recognize which digit it shows?

Accuracy is not the whole story

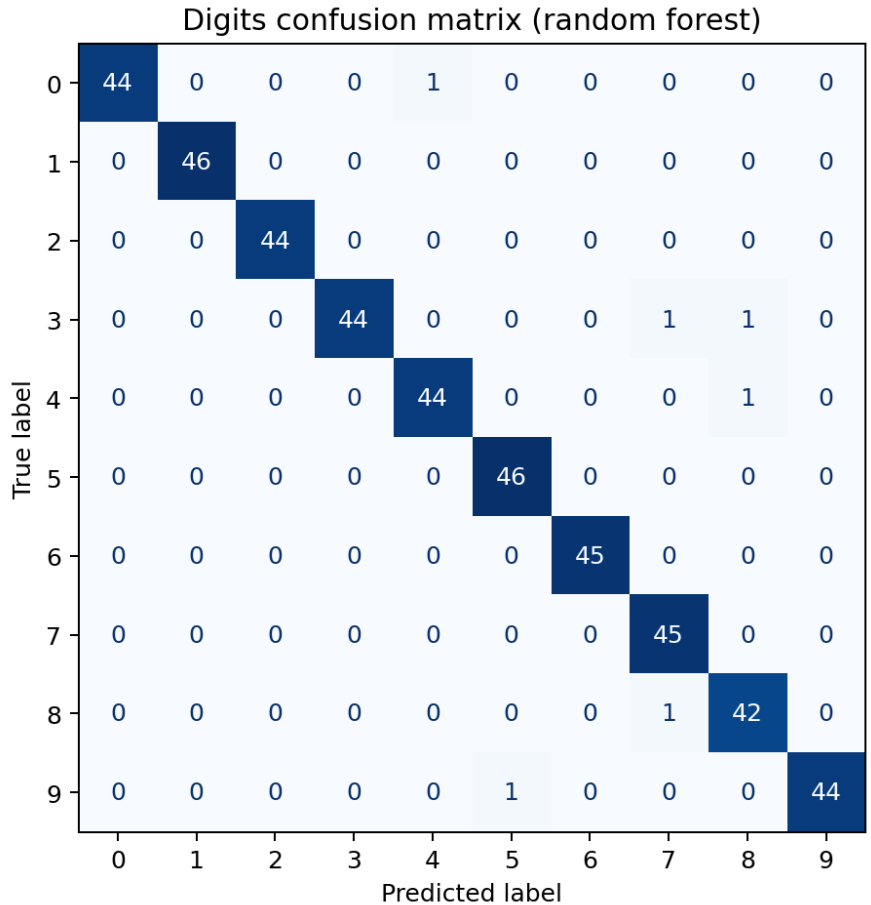
Accuracy tells us the share of all predictions that are correct.
It does not reveal which digit pairs are confused.

A **confusion matrix** shows how actual and predicted classes compare.

- The diagonal shows correct predictions.
- Off-diagonal counts show specific mistakes.

A useful model report combines an overall metric with an error analysis.

Look for the mistakes



Find a cell outside the diagonal.
Which true digit was predicted as which other digit?

Which claim is more defensible?

Claim A: A random forest is always better than a single tree.

Claim B: A random forest may generalize better, but the choice depends on the data, the evidence, and the need for interpretability.

Choose a claim; provide one reason and one piece of evidence you would want (3 min).

Your final in-class assignment

Question: Should we use a decision tree or a random forest to classify the handbook's handwritten digits?

Deliverable: One runnable notebook, plus a 150–250 word recommendation inside it.

Expectation: Make a fair comparison, interpret the mistakes, and defend a choice.

> Use the held-out test set for your final comparison, not for repeated model tuning.

Assignment: what to include

1. Load `load_digits()` and describe the data.
2. Use one stratified 75/25 split (`random_state=0`).
3. Fit a tree and a forest on the same training set.
4. Report test **accuracy**, **macro-F1**, and a **confusion matrix** for each.
5. Identify one important error pattern or tradeoff.
6. Recommend one model and acknowledge one limitation.

Start your notebook here

```
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (accuracy_score, f1_score,
                             confusion_matrix)

X, y = load_digits(return_X_y=True)
Xtr, Xte, ytr, yte = train_test_split(
    X, y, test_size=.25, stratify=y, random_state=0)

# TODO: fit both models, predict, and compare
# TODO: report accuracy, macro-F1, confusion matrices
# TODO: recommend a model and state one limitation
```

A simple plan for the class period

- **0–5 min:** Inspect the data and predict the outcome.
- **5–15 min:** Fit the two models.
- **15–25 min:** Compute metrics and study errors.
- **25–35 min:** Write the recommendation.
- **35–40 min:** Exchange notebooks for peer review.

Prioritize a reproducible, well-explained comparison over visual polish.

Holistic rubric: one score from 0 to 4

4 — Excellent Fair, reproducible comparison; insightful error analysis; convincing recommendation and meaningful limitation.

3 — Proficient Both models compared correctly; reasonable recommendation; minor gaps in explanation or reproducibility.

2 — Developing Basic comparison present, but important gaps in evaluation or reasoning.

1 — Beginning Incomplete or substantially flawed comparison, with little useful interpretation.

0 — No evidence No assessable comparison submitted.

Score the whole notebook, not separate checklist items.

Peer review and final reflection

Ask a peer:

- Does the notebook run from top to bottom?
- Are both models evaluated on the same held-out data?
- Does the recommendation match the evidence?
- Which rubric score best describes the whole submission?

Complete: “I would choose . . . because . . . . The strongest evidence is . . . . My biggest uncertainty is . . . .”

A decision tree makes predictions through a series of questions.

A random forest builds many different trees and combines their predictions.

Neither model is automatically best: **the right choice depends on fair evaluation and the task.**